Final Project

Start Assignment

- Due Nov 18 by 10pm
- Points 70
- Submitting a text entry box
- Available until Nov 18 at 11:30pm

Assignment

Demonstrate your ability to use, evaluate, and govern GenAI in an applied analytic context. **Work in pairs** (assigned/formed based on the Week 0 survey) - one shared workflow, one shared submission, with individual accountability built in where it matters most (see AI use log below). You will also use a single GitHub repo to house your project.

Submission components

One submission per pair unless noted otherwise:

1. **Reproducible Python notebook** - Submit as a Jupyter notebook (`.ipynb`) rendered as `.html` that runs top to bottom without errors. Runs end-to-end with assertions, documented cleaning, and formal evaluation metrics. Associated GitHub repo is available to look at.

2. **Written summary (3-5 pages)** - Submit as a PDF. Required sections:

- Introduction (use case, question, significance)
- Data (source, size, unit of observation, summary statistics)
- Methods (approach justification, AI tools and settings, evaluation approach, assumptions)
- Results (findings with effect sizes, CIs, P/R/F1, kappa, cost analysis - as applicable)
- Governance (risk level, key rules, quantitative thresholds - a short summary; see Component 3 for the full memo)
- Reflection (what you each learned, what you'd change)

3. **Governance memo** - Submit as a separate PDF from your written summary. Built from your two individual Week 9 governance memos, reconciled and revised for your shared final workflow, with a quantitative risk matrix and measurable thresholds. It's the full memo - more detailed than the Governance section in Component

4. **Presentation slides** - Submit as a PDF export of the slides you present in Week 10.

5. **GitHub repo -** One repo per pair holding all of your Python code (`.ipynb` and any `.py` files), a README, and everything needed to run it. Submit the link on Canvas. See the GitHub Repo Guidelines page on Canvas for what to include.

6. **AI use log** - **submitted individually, one per partner.** Each of you documents your own AI interactions, judgment calls, and verification. This is the one component that isn't shared because it's meant to reflect your personal decisions, not a joint narrative. See the **AI use log guide** (https://steflangehennig.github.io/info4340/guides/ai-use-log/) .

Presentation

10 minutes per pair: 8 presenting + 2 Q&A. Split speaking time so both partners present part of the work. Demonstrate all four competencies (understand, use, evaluate, govern) with quantitative evidence. Q&A will include at least one question directed at each partner individually, so come ready to speak to the whole project, not just your half.

Peer evaluation

During Week 10, you'll also score 2-3 other pairs' presentations using a simplified rubric (workflow clarity, evaluation rigor, governance substance, presentation quality), submitted via Canvas the same day. This is a separate task, not part of your own submission. Your presentation will also be scored by peers as one input alongside grading.

Rubric

The first six criteria are scored once per pair - both partners receive the same score. **AI use log is scored individually** since it's submitted separately by each partner.

Criterion	Excellent (10)	Adequate (6)	Needs revision (3)
Use case and workflow	Clear, realistic with well-designed pipeline	Identified but incomplete	Vague or no pipeline
Reproducible notebook	Runs end-to-end, assertions, formal evaluation, cleaning documented	Runs but sparse documentation	Doesn't run or major gaps
GitHub repo	All code, README with every section and real findings, `requirements.txt` , data or access instructions, saved AI outputs; runs from a fresh copy; no API keys committed	Code and README present, but README is missing sections or setup steps are incomplete	Missing code or README, or can't be run from a fresh copy
Written summary with methods	All 6 sections, formal methods, summary stats, metrics with CIs	Most sections but methods lack rigor	Missing sections or no methods
Governance memo	Risk matrix, 3+ rules with measurable thresholds, escalation	Some elements with some thresholds	Principles only
Presentation	All four competencies, quantitative evidence, engages audience; both partners speak substantively	Most elements but uneven, or one partner dominates	Missing major components
AI use log (individual)	Comprehensive, shows evolving usage and judgment	Present but incomplete	Missing

The total is 70 points per student - each partner is graded individually on the AI use log row and shares the same score on the other six.

Final Project

Criteria	Ratings				Pts
Use case and workflow Use case and workflow	**10 pts** **Excellent** Clear, realistic with well-designed pipeline	**6 pts** **Adequate** Identified but incomplete	**3 pts** **Needs revision** Vague	**0 pts** **No Evidence** No pipeline	10 pts
Reproducible notebook + GitHub repo Reproducible notebook + GitHub repo	**10 pts** **Excellent** Runs end-to-end, assertions, formal evaluation, cleaning documented	**6 pts** **Adequate** Runs but sparse documentation	**3 pts** **Needs revision** Doesn't run or major gaps	**0 pts** **No Evidence** Missing	10 pts
GitHub repo GitHub repo	**10 pts** **Excellent** All code, README with every section and real findings, requirements.txt, data or access instructions, saved AI outputs; runs from a fresh copy; no API keys committed	**6 pts** **Adequate** Code and README present, but README is missing sections or setup steps are incomplete	**3 pts** **Needs revision** Missing code or README, or can't be run from a fresh copy	**0 pts** **No evidence** Missing	10 pts
Written summary with methods Written summary with methods	**10 pts** **Excellent** All 6 sections, formal methods, summary stats, metrics with CIs	**6 pts** **Adequate** Most sections but methods lack rigor	**3 pts** **Needs revision** Missing sections	**0 pts** **No Evidence** Missing/no methods	10 pts
Governance memo Governance memo	**10 pts** **Excellent** Risk matrix, 3+ rules with measurable thresholds, escalation	**6 pts** **Adequate** Some elements with some thresholds	**3 pts** **Needs revision** Principles only	**0 pts** **No Evidence** Missing	10 pts
Presentation Presentation	**10 pts** **Excellent** All four competencies, quantitative evidence, engages audience; both partners speak substantively	**6 pts** **Adequate** Most elements but uneven, or one partner dominates	**3 pts** **Needs revision** Missing major components	**0 pts** **No Evidence** Completely missing	10 pts
AI use log (individual) AI use log (individual)	**10 pts** **Excellent** Comprehensive, shows evolving usage and judgment	**6 pts** **Adequate** Present but incomplete	**0 pts** **Needs revision** Missing		10 pts

Course Chat